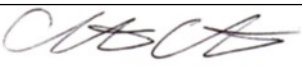


Guideline Title: Non-Invasive Positive Pressure Ventilation

Guideline Number: GUID-3203-PR018

Issue date: 09/09/2014	Replaces Dept. Policy:
Revision dates: 4/30/24	Developed by: Air Care Team
Approved by: Dr. Christopher Hunter, MD Air Care Team Medical Director	Approved by:
Signature: 	Signature:
Department Numbers: 3203	

- I. **PURPOSE:** To provide adequate support and oxygenation to patients not yet requiring intubation and invasive mechanical ventilation.
- II. **DEFINITIONS:**
 - a. BiPAP – Bilevel Positive Airway Pressure
 - b. EPAP - Expiratory positive airway pressure
 - c. NPPV – Non-Invasive Positive Pressure Ventilation
 - d. CPAP – Continuous Positive Airway Pressure
 - e. IPAP - Inspiratory positive airway pressure
 - f. PEEP – Positive end-expiratory pressure
- III. **DEPARTMENT GUIDELINE:**
 - a. Types of NPPV
 - i. CPAP – Continuous positive airway pressure mode – provides continuous PEEP
 1. Indications: congestive heart failure (CHF), pulmonary edema, obstructive sleep apnea
 - ii. BiPAP – Bi-level positive airway pressure mode – applies a continuous PEEP throughout the breath cycle, but also pressurizes to a set pressure during inhalation
 1. Indications: Asthma, chronic obstructive pulmonary disease (COPD), Resp failure from exhaustion, CHF, or pulmonary edema
 - b. **EQUIPMENT:**
 - i. NPPV mask from sending facility
 - ii. Oxygen source
 - iii. Revel transport ventilator
 - iv. Disposable ventilator tubing
 - c. **INDICATIONS:**
 - i. Any patient requiring ventilatory or oxygenation support who does not yet require intubation or refuses intubation on an interfacility transport.
 - ii. Hypoxemia secondary to pulmonary edema
 - d. **CONTRAINDICATIONS:** Penetrating chest trauma, persistent nausea/vomiting, obtundation, respiratory arrest, or inability to protect patient's airway
 - e. **PROCEDURE:**
 - i. Assess/manage airway, breathing and circulation. Assess ECG, SpO2 and blood pressure.
 - ii. Explain the procedure to the patient

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- iii. Fit and size mask to the patient. Prepare the headpiece and circuit according to manufacturer's specifications. Ensure mask vents are CLOSED.
 - iv. Attach gas outlet, pressure transducer, & exhalation valve tubes to corresponding connectors.
 - v. Turn the ventilator on allow self-check to perform.
 - vi. Select appropriate patient size, scroll to NPPV, and hit select.
 - vii. Initiate settings appropriate for condition or begin with referring hospital settings.
 - 1. BiPAP (Target initial is IPAP/EPAP of 10/5 cm/H₂O). Pressure support is the difference between IPAP and EPAP.
 - 2. CPAP (Target initial is IPAP/EPAP 0/ 5 cm/H₂O)
 - 3. If transferring from an open to closed system, drop IPAP by 4 5 cm/H₂O (but not less than 10 5 cm/H₂O total IPAP), dial up as necessary.
 - viii. Use appropriate PEEP setting, ensuring correlation between tubing and desired PEEP level.
 - ix. Attach ventilator to 50psi oxygen source utilizing high pressure tubing. Ensure that the portable oxygen tank has a minimum of 1000 PSI of pressure or connect oxygen tubing to oxygen inlet 15 lpm flow and dial the FiO₂ to read the low-pressure source.
 - x. Set breath rate to 0 or consider a rate to augment the breath support.
 - xi. Set FiO₂. Titrate to use lowest concentration.
 - xii. Turn off low pressure and minute ventilation alarms, set appropriate for patient.
 - xiii. Attach ventilator circuit to the mask and administer treatment.
 - xiv. Troubleshooting. If patient does not tolerate ventilator settings...
 - 1. Enter the ventilator's extended features by pushing and holding the monitor "SELECT" key for three (3) seconds and scroll to 'Vent Control' menu
 - a. Adjust rise profile to 1-2. A faster rise time will deliver O₂ quickly.
 - b. Adjust flow termination to 40% to prevent prolonged inhalation.
 - c. Adjust time termination to 0.7
 - 2. If SpO₂ remains low, increase the EPAP or FiO₂.
 - 3. If ETCO₂ low, reduce the pressure support, thus lowering IPAP
 - 4. If ETCO₂ high, increase the pressure support (PS), thus elevating IPAP
 - 5. Exit the extended features menus by turning the set value knob until exit is displayed and push Select until monitored data is displayed in the window.
 - xv. If patient is agitated, coach patient to breathe calmly and consider light sedation
 - xvi. Consider a 30 second trial of the patient lying flat prior to heading aircraft to assess if patient will tolerate lying flat while being loaded into the aircraft.
 - xvii. Continue treatment during the entire transport.
 - xviii. If patient is receiving a nebulized bronchodilator, it may be continued by placing the nebulizer device in-line with the NPPV circuit.
 - xix. Reassess vital signs every five minutes while the patient is on NPPV and document.
 - xx. If respiratory status deteriorates, consider intubation or other airway management.
- f. **COMPLICATIONS:** Hypotension or pneumothorax

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- IV. **DOCUMENTATION:** In EMS Charts, document mode of ventilation & all ventilator settings (the rise profile, flow term, and time term). Document measurements of ventilation (Vte, VE, and frequency).
- V. **REFERENCES:**